

Does the introduction of the 2016 UK National Living Wage (NLW) increase wages and affect employment differently across regions with varying exposure to the policy?

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Econometrics and Data Science Empirical Project

Student ID: 11301282, 11298230, 11360410, 11011179

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1 Introduction

The implementation of the 2016 UK National Living Wage (NLW) generated debate over whether national wage floors reduce regional inequality or trigger disemployment. We employ a Difference-in-Differences (DID) framework using ASHE and LFS data (2014–2018) to investigate whether a national wage floor can improve living standards in lagging regions without undermining local job prospects. Specifically, we examine if high-exposure regions saw significant differential gains in earnings and employment relative to low-exposure areas. Findings indicate the NLW raised earnings generally, but failed to narrow regional disparities. Crucially, we find no evidence of cross-regional adverse employment effects.

2 Literature Review

Research on income inequality and the associated policy levers often focuses on uplifting low-paid workers and/or factors that affect productivity.

Stewart (2004) establishes the methodological template. Through wage group Difference-in-Difference, they compare workers directly affected by increases to National Minimum Wage (NMW) to workers just above the incoming threshold (treatment versus control). Dickens et al. (2015) refined the approach, concluding that the NMW reduced employment retention among part-time female workers. Fidrmuc and Tena (2018), however, propose opposition to the parallel trends assumption. They use RDD to find that minimum wage may have smoother effects, with anticipatory behaviour across low and high-bite age groups. This is the motivation to separate employment and age in our examination.

Moving closer to studies on the NLW, precedent is established by Aitkens et. al (2017), employing a Difference-in-Difference approach but with synthetic control methods. They find that the NLW uprating in 2017 had (on a national-level) minimal employment retention effects. While Dube (2019) provides comprehensive national evidence, Butcher and Dickens (2023) apply a more relevant geographic DID on region-age-gender cells. They

found that those most affected by the NLW are those that saw faster average wage increases (Butcher and Dickens, 2023). In approach, our exposure measure is instead binary and collects aggregate-age, region-level data.

3 Data

3.1 Data Construction and Cleaning

Our study combines two publicly available UK labour market datasets to examine regional variation in the effects of the 2016 National Living Wage (NLW). Firstly, the Annual Survey of Hours and Earnings (ASHE) provides annual regional earnings data. From the raw ASHE files across 2014-2018, “All employees” tables were extracted and cleaned to retain only region-level observations, removing age-specific breakdowns and aggregate UK figures. “Median hourly pay excluding overtime” is used as the relevant wage measure, as it provides a robust indicator of typical earnings. The annual datasets were combined to form a balanced region-year panel covering 12 UK regions over five years, yielding 60 effective observations.

Table 0: Variable Definitions and Data Sources

Variable	Description	Source	Units
Hourly_Pay	Median hourly pay for all employees in the region	ASHE (ONS)	GBP (£)
employment_rate	Percentage of the population aged 16-64 in employment	APS (ONS)	Percentage (%)
HighExposure	Binary indicator: 1 if the regional median wage was below the pre-treatment cross-regional mean (2014–2015)	Computed	Binary (0/1)
Post	Binary indicator: 1 for periods after the NLW introduction (2016+)	Computed	Binary (0/1)
Year	Year of observation (annual frequency)	ONS	Year

Secondly, the Labour Force Survey / Annual Population Survey (LFS/APS) provides regional high-frequency measures of employment. From the “People” sheets, “Employment rate (%) for individuals aged 16-64” was selected as the relevant variable. Monthly observations were restricted to 2014-2018 and averaged within each year to produce a consistent annual measure for each region. This produced a second balanced region-year panel with an identical structure, ensuring comparability with wages.

3.2 Final Dataset and Variable Definition

To capture variation in policy exposure, regions are classified according to pre-policy wage levels, with those below the 2014-15 overall average median wage (£11.39/hr) defined as ‘high-exposure’ to the NLW. Each region’s 2014-15 average, and corresponding ‘exposure’ status was then reported. This reflects differential exposure, as lower wage regions are more likely to have a higher share of workers directly affected by the policy. The final dataset was constructed by merging the aforementioned wage and employment panels, using common region and year identifiers, resulting in a region-year panel of the working-age population. The dataset includes the key outcome variables (median hourly pay and employment rate), alongside the treatment indicator (high-exposure) and post-policy dummy, forming the basis for the difference-in-differences approach.

Table 1: Descriptive Statistics

Statistic	N	Mean	St. Dev.	Min	Max
Hourly Pay (£)	60	11.81	1.53	10.12	17.11
Employment Rate (%)	60	73.63	2.99	67.93	79.20
High Exposure Dummy	60	0.75	0.44	0	1

3.3 Data Limitations

Both survey-based datasets rely on weighting procedures to be entirely representative of the UK; this may introduce measurement error regionally. Furthermore, constructing annual averages from monthly rolling observations may smooth short-term variation and attenuate immediate policy effects.

The relatively small sample size (60 observations) and limited control group (three ‘low-exposure’ regions: London, Scotland, South-East) may reduce statistical power and the precision of estimated effects, increasing the risk of a Type II error and leaving results sensitive to idiosyncratic shocks in the South East and London.

4 Methodology

We identify the causal impact of the 2016 NLW using a Difference-in-Differences (DiD) framework. Exploiting regional variation in bite, this design compares high-exposure regions against low-exposure controls over time to isolate the policy effect from broader macroeconomic trends.

4.1 Empirical Specification and Model Setup

Following Butcher and Dickens (2023), we estimate the NLW’s causal effect on regional earnings and employment using the following baseline DID model:

$$Y_{rt} = \alpha + \beta_1 \text{HighExposure}_r + \beta_2 \text{Post}_t + \beta_3 (\text{HighExposure}_r \times \text{Post}_t) + \epsilon_{rt}$$

Where:

- Y_{rt} : The outcome of interest (median hourly pay excluding overtime or the employment rate for individuals aged 16-64) for region r in year t

- HighExposure_r : A binary variable identifying regions with pre-policy median pay below the £11.39 average
- Post_t : A temporal dummy equal to 1 for years ≥ 2016
- β_3 : Coefficient of interest, representing the Difference-in-Differences estimate

4.2 Robustness: Two-Way Fixed Effects (TWFE)

To control for unobserved heterogeneity, we estimate a TWFE specification:

$$Y_{rt} = \beta_3(\text{HighExposure}_r \times \text{Post}_t) + \gamma_r + \delta_t + \epsilon_{rt}$$

Region fixed effects (γ_r) control for permanent regional differences, such as historical productivity, while year fixed effects (δ_t) absorb national shocks. In this model, the individual treatment and time dummies are accounted for, isolating β_3 as the specific measure of the policy’s causal impact.

4.3 Identification Logic and Assumptions

For β_3 to be interpreted as a causal effect, the parallel trends assumption must hold: in the absence of the NLW, high and low-exposure regions would have followed comparable trajectories. We support this via placebo regressions (see Table 3) on the pre-reform period (2014-2015). Insignificant interaction terms during these years establishes a robust counterfactual. Furthermore, for β_3 to represent a strictly causal effect, these conditions must hold:

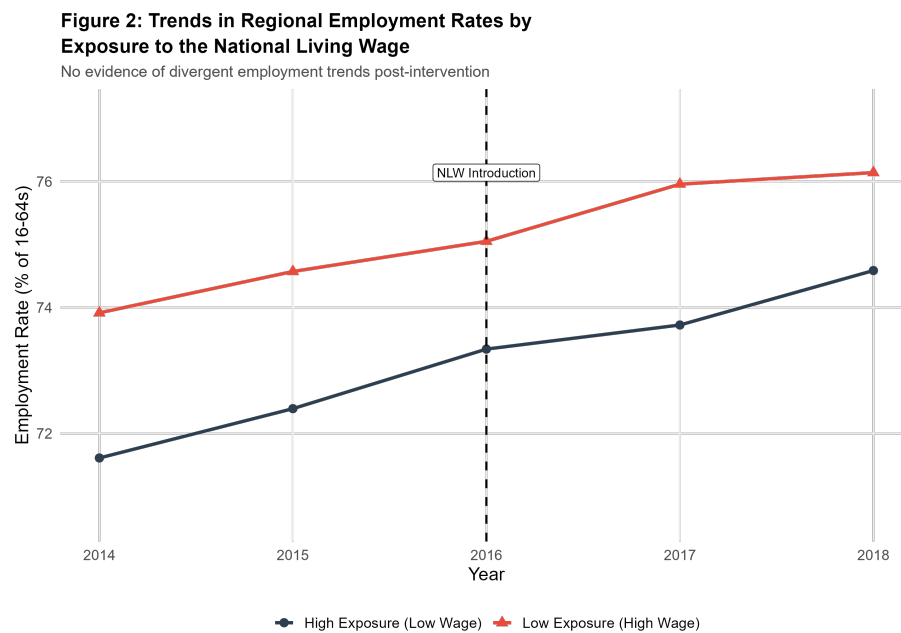
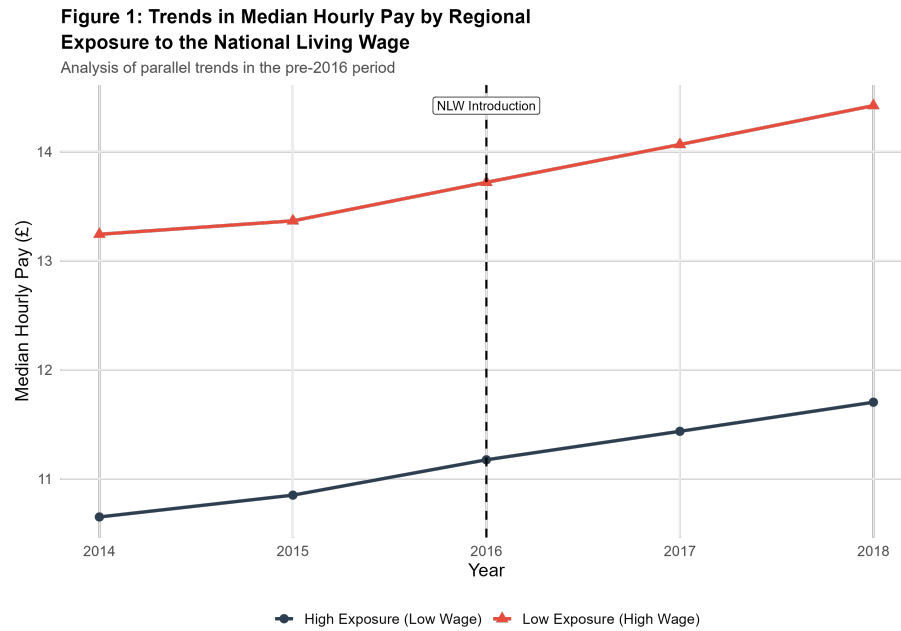
1. **No Concurrent Shocks:** No other localised policies or industrial shifts occurred in April 2016 that affected low-wage regions differently than high-wage regions.
2. **No Spillover Effects (SUTVA):** There was no significant labour mobility across regional borders (e.g., workers migrating from the North to London) in direct response

to the wage change.

4.4 Limitations and Threats to Identification

1. **Spatial Aggregation Bias:** NUTS1 regions are geographically broad. Significant shifts in specific cities might be averaged out across the wider region, potentially making the policy's impact appear smaller than it is at the local level.
2. **Demographic Substitution:** As the NLW only affected those aged 25+, aggregate data (16-64) may mask labour-labour substitution. Firms may replace older workers with exempt younger staff, leaving total employment rates unchanged.
3. **Measurement of Bite:** Regional averages fail to capture the specific density of low-wage earners. This lack of distributional precision likely introduces attenuation bias, potentially negatively biasing our interaction coefficients.
4. **Time Horizon:** The 2014-2018 window captures immediate adjustments but misses long-term structural shifts, such as firm-level investment in automation or capital-labour substitution.

5 Results



Figures 1 and 2 illustrate the trends in median hourly pay and employment rate across high and low exposure regions before and after the introduction of the NLW. In both cases, the groups follow similar trajectories between 2014 and 2015, visually supporting the parallel

trends assumption. Following the 2016 intervention, both groups continue to move in tandem, suggesting that the NLW did not induce treatment and control group divergence in either case.

Table 2: Impact of the 2016 National Living Wage on Regional Outcomes

	<i>Dependent variable:</i>			
	Hourly Pay (£)		Rate (%)	
	Wages (1)	Wages (2)	Employment (3)	Employment (4)
High Exposure Region	-2.554*** (0.465)	-1.085*** (0.073)	-2.241* (1.315)	-0.690 (0.426)
Post-2016 (NLW)	0.765 (0.520)	1.142*** (0.061)	1.473 (1.470)	2.483*** (0.357)
DiD (High Exposure x Post)	-0.078 (0.600)		0.407 (1.697)	
HighExposure:Post		-0.078 (0.061)		0.407 (0.355)
Constant	13.305*** (0.403)	12.259*** (0.058)	74.244*** (1.139)	76.586*** (0.341)
Region Fixed Effects	No	Yes	No	Yes
Year Fixed Effects	No	Yes	No	Yes
Observations	60	60	60	60
R ²	0.605	0.997	0.173	0.972
Adjusted R ²	0.584	0.996	0.128	0.962

Note: Standard errors in parentheses. *p<0.05, **p<0.01, ***p<0.001.

5.1 Impact on Regional Wages

In the TWFE model, the Difference-in-Differences (DiD) estimator for hourly wages is -0.078 (SE = 0.061, p = 0.207). This negative coefficient suggests a slowing of wage growth in high exposure regions but is economically negligible and is not statistically significant. The coefficient is stable across OLS and TWFE. This consistency suggests a robust null result that the wage gap remained unchanged. The high R^2 value (0.99) reflects the high persistence in regional labour market characteristics, with TWFE absorbing most variation in wage levels and employment rates.

5.2 Impact on Employment

Neoclassical economic theory would predict that if the NLW were set above the marginal productivity of labour in low-wage regions, we would expect a relative contraction in employment. However, our TWFE estimate (Column 4) yields a DiD coefficient of 0.407 (SE = 0.355, $p = 0.811$).

This positive coefficient is contrary to disemployment fears. However, it is small relative to the employment rate and statistically insignificant. Across OLS and TWFE models, this coefficient is stable at 0.407, adding robustness to the null of no effect of NLW on relative employment rates. This absorption of wage costs likely occurred through channels other than headcount reduction, such as efficiency wage effects or price pass-through to consumers.

5.3 Internal Validity and Parallel Trends

Table 3: Placebo Test for Parallel Trends (2014-2015)

	<i>Dependent variable:</i>	
	Hourly Pay (£) Wages (1)	Rate (%) Employment (2)
High Exposure Region	-1.248*** (0.104)	-0.065 (0.487)
Fake Post (2015)	0.123 (0.077)	0.656* (0.359)
Placebo Interaction (Trend Test)	0.077 (0.089)	0.127 (0.415)
Constant	12.313*** (0.077)	76.414*** (0.359)
Region Fixed Effects	Yes	Yes
Observations	24	24
R ²	0.998	0.991
Adjusted R ²	0.996	0.979

Note:

Standard errors in parentheses. All coefficients are statistically insignificant at the 5% level, supporting the parallel trends assumption.

Table 3 presents the results of a placebo test, using 2015 as the fake treatment year. The resulting interaction term for wages is 0.077 (SE = 0.089, $p = 0.408$) and for employment is 0.482 (SE = 0.414, $p = 0.252$). The failure to reject the null hypothesis of zero differential

trends in the pre-reform period provides strong empirical support for the parallel trends assumption used in the DiD regressions in Table 2.

6 Conclusion

This study examined whether the 2016 National Living Wage had differential effects on wages and employment across UK regions with varying exposure. Results showed no statistically significant difference in either outcome between high and low exposure regions, suggesting the policy did not disproportionately affect more exposed regions. While the results indicated no evidence of adverse macroeconomic effects, conclusions are limited by a short time horizon and aggregate regional data. Future work could explore more granular or longer-term impacts.

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Appendix: Data Construction

Data Sources

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Office for National Statistics (ONS) (2025) Headline labour force survey indicators for all regions (HI00). Available at: <https://www.ons.gov.uk/employmentandlabourmarket/peopleinwork/employmentandemployeetypes/datasets/headlinelabourforcesurveyindicatorsforallregionshi00> (Data used: "employment rate (%) for individuals aged 16-64", annualised from monthly data, 2014-2018)